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Ten
Beta Beyond ~~Twenty~~

Skin and extremity dose rates from external beta sources not close to the skin

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44 slides:
Why: How: Where

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Meeting IRR17 requirements

if you have a beta source, you'll probably want to *anticipate* likely skin and extremity dose rates

- (reg 8) as part of your initial risk assessment
- (reg 8) modelling foreseeable accidents
- (reg 9) quantitatively comparing alternative arrangements
- (reg 24/26) retrospective dose reconstruction
- ... and others.

Varskin is the go-to for close skin contact, but what about > 20 cm away

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How: IRR17 doesn't specify any dose measurement/estimation physics

ICRP: The equivalent dose at 0.07 mm skin depth averaged over any 1 cm² of the skin, regardless of the area exposed. **But not directly measurable.**

IRR17 guidance says the measurable stand-in, ICRU Hp(d), can be used to demonstrate compliance with dose limits. (L121 Appendix 2)

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So what's available for straightforward beta-at-a-distance skin-dose prediction [Hp(0.07)] ?

1. **$D \approx 7.5 \times 10^{-12} A/d^2$ Gy/hr** (d in m, A in Bq). Quick & dirty absorbed dose rates, a 1970s-vintage precursor to Hp(0.07), an energy-independent approximation
2. **Delacroix at 30 cm**, inverse square law for other distances. Based on Varskin (1992) (known to be iffy). Stated as Hp(0.07)
3. **Online calculators**. Commonly traced to Cross *et al.* If it's Cross *et al.* (1982), these are not actually Hp(0.07) estimates. They are absorbed-dose rates calculated for homogeneous media.

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How good are these prediction methods?

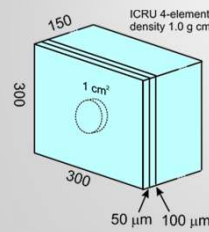
The benchmark is Monte Carlo modelling using a reference computational tissue slab, a computational 'phantom'.

1. There is more than one reference computational phantom!
2. Monte Carlo modelling is not 'plug and play'. Probably quicker to solve cold nuclear fusion in your garage.....

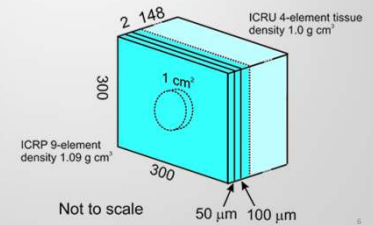
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ICRU57 Skin tissue computational phantom



ICRU95/ICRP Skin tissue (trunk) computational phantom



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Common nuclear particle transport MC toolkits (n.b. they're not standalone programs)

MCNP6 - Los Alamos National Laboratory, United States. Export-controlled. DOS-like.

EGSnrc - National Research Council Canada. Open source, has to be compiled.

Geant4 - International Geant4 Collaboration group. Open source, hosted by CERN. It's a C++ based toolkit.

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Comparison with MCNP6.2/ICRU 95 skin tissue phantom

10 MBq P-32 point source dose rate ($\mu\text{Sv/h}$) (Emax 1.7107 MeV)
[Dose: Hp(0.07)]

Distance /cm	MCNP6.2 & ICRU95 phantom	Amato et al (Geant4 & ICRP)	Varskin+2.1	Delacroix 2024	Cross et al	Rule of Thumb
10	12860	12800	7300		9300	7500
20	3253		1900		2350	1875
30	1480	1430		1200	1050	833
40	833				595	469
50	500	507			380	300
60	352				266	208



Each MCNP run was 1E8 'histories' and took 80 minutes per run

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Comparison with MCNP6.2/ICRU 95 skin tissue phantom

10 MBq Sr-90 point source dose rate ($\mu\text{Sv/h}$) (Emax 0.546 MeV)
[Dose: Hp(0.07)]

Distance /cm	MCNP6.2 & ICRU95 phantom	Amato et al (Geant4 & ICRP)	Varskin+2.1	Delacroix 2024	Cross et al	Rule of Thumb
10	16962	16300	11000		13469	7500
20	3746		2300		2901	1875
30	1454	1240		960	1106	833
40	650				521	469
50	325	275			271	300
60	178				147	208

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Comparison with MCNP6.2/ICRU 95 skin tissue phantom

10 MBq P-33 point source dose rate ($\mu\text{Sv/h}$) (Emax 0.2485 MeV)
[Dose: Hp(0.07)]

Distance /cm	MCNP6.2 & ICRU95 phantom	Petoussi et al 1993 H'(0.07,0)	Varskin+2.1	Delacroix 2024	Cross et al	Rule of Thumb
10	7852	7350	5200		11000	7500
20	787		290		1037	1875
30	75			0	107	833
40	11				6	469

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Conclusion - they are all iffy. They mostly underestimate based on M.C./ICRU95, often by well over 40%

Which is often all you need, eg a ball-park figure.

But - sometimes you need better when it comes to:

- calculating what you need for shielding
- evaluating different working arrangements
- retrospective dose estimations
- .. and so on

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A proposal for an easy-to-use but more reliable algorithm (say aspiration of $\sim 10\%$ within M.C. estimates)

Multiply the fluence rate at the skin surface by the fluence-to-dose coefficient

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B. Break it down into 6 steps

Calculate the geometry factor into 1 cm² of skin at x cm

For a narrow energy interval, estimate the electron energy remaining at the skin surface

Use the fluence-to-dose coefficient to find the absorbed dose of the electron into 1 cm² of skin

Multiply by the number of electrons/second incident on the 1 cm² of skin

Do that for all the beta energy spectrum

Add up all the energy-interval absorbed dose rates

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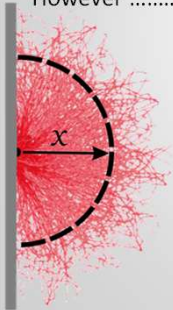
Step 1. Calculate the geometry factor

Electrons have tortuous paths

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However



Worst beta deflections happen late.

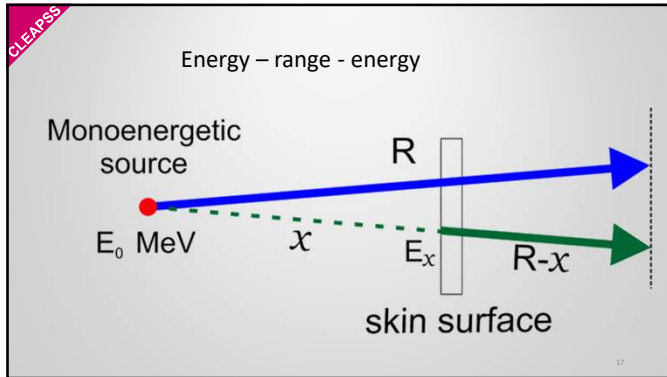
So for monoenergetic electrons, if you are not near the range end, the geometry factor for a one sq cm area at x metres $\approx \frac{1}{4\pi x^2}$

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Step 2. For a narrow energy interval, estimate the electron energy remaining at the skin surface

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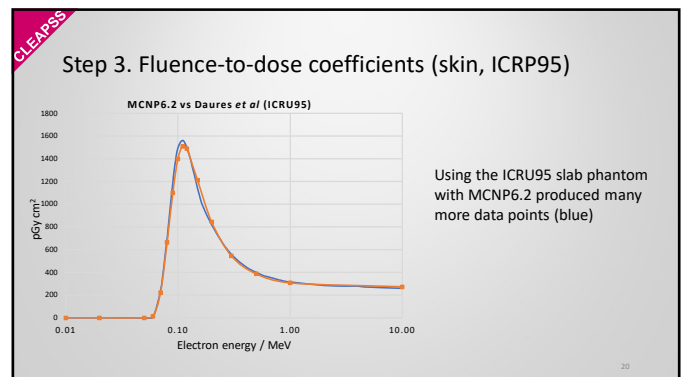
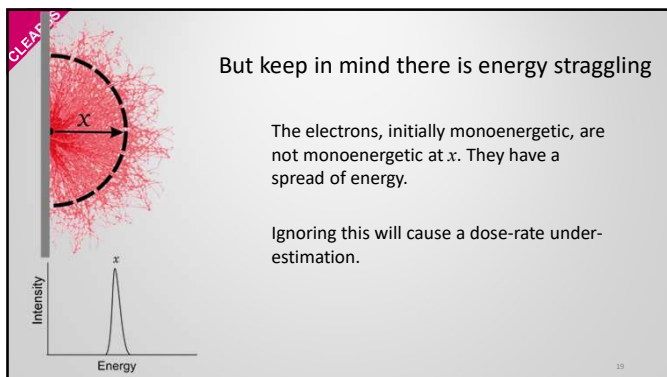
Katz & Penfold range equations are well-established and turn out to be invertible

$$R = 0.412 E^{1.265 - 0.0954 \ln E} \quad R = 0.530E - 0.106$$

$E < 2.5$ MeV $E \geq 2.5$ MeV

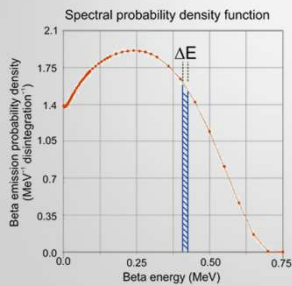
$$E = \begin{cases} \exp \left[\frac{1.265 - \sqrt{1.600225 - 0.3816 \ln(R/0.412)}}{0.1908} \right] & R < 1.219 \text{ g cm}^{-2} \\ \frac{R + 0.106}{0.530} & R \geq 1.219 \text{ g cm}^{-2} \end{cases}$$

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Step 4. Multiply by the number of electrons/second incident on the 1 cm² of skin – ICRP-07.BET



From ICRP-107. Spectral probability density functions for 955 beta radionuclides

The probability that a nuclear disintegration produces a beta particle with an initial energy in the range shown by ΔE is the area of the shaded part under the graph.

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Steps 5&6. Put it all together

Absorbed dose rate =

the probability of beta emission at a specific energy interval

x activity

x geometric factor

x residual energy at the skin

x fluence to dose coefficient for that specific energy interval
summed over all the energy intervals

And for electrons, an absorbed dose of 1 Gy into the skin at 50 μm depth in a layer 50 μm thick leads to personal dose equivalent Hp(0.07) of 1 Sv

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But what about the energy straggle?



Turns out that for a wide range of beta radionuclides, ignoring the energy straggling caused a remarkably consistent under-estimation by ~ 10% for point sources.

An (optional) empirical energy straggling factor was used to account for this.

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BBT results - comparison

10 MBq P-32 point source dose rate ($\mu\text{Sv/h}$) (Emax 1.7107 MeV)
[Dose: Hp(0.07)]

Distance /cm	MCNP6.2 & ICRU95 phantom	Beta Beyond Ten	Amato et al (Geant4 & ICRP)	Varskin+2.1	Delacroix 2024	Cross et al	Rule of Thumb
10	12860	13085	12800	7300		9300	7500
20	3253	3241		1900		2350	1875
30	1480	1401	1430		1200	1050	833
40	833	788				595	469
50	500	484	507			380	300
60	352	336				266	208

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BBT results - comparison

10 MBq Sr-90 point source dose rate ($\mu\text{Sv/h}$) (Emax 0.546 MeV)
(Normally incident) Hp(0.07)

Distance /cm	MCNP6.2 & ICRU95 phantom	Beta Beyond Ten	Amato et al (Geant4 & ICRP)	Varskin+2.1	Delacroix 2024	Cross et al	Rule of Thumb
10	16962	17055	16300	11000		13469	7500
20	3746	3653		2300		2901	1875
30	1454	1362	1240		960	1106	833
40	650	662				521	469
50	325	336	275			271	300
60	178	192				147	208

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BBT results – comparison

10 MBq P-33 point source dose rate ($\mu\text{Sv/h}$) (Emax 0.2485 MeV)
(Normally incident) Hp(0.07)

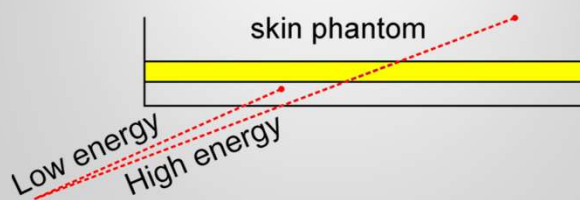
Distance /cm	MCNP6.2 & ICRU95 phantom	Beta Beyond Ten	Varskin+2.1	Delacroix 2024	Cross et al	Rule of Thumb
10	7852	7359	5200		11000	7500
20	787	607	290		1037	1875
30	75	49		0	107	833
40	11	0.2			6	469

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Extending to disc sources

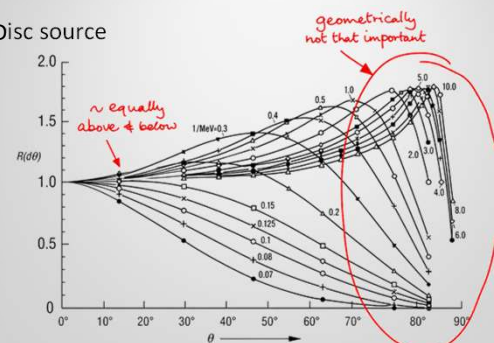
Fluence-to-dose coefficient is affected by the angle of beta incidence



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Disc source



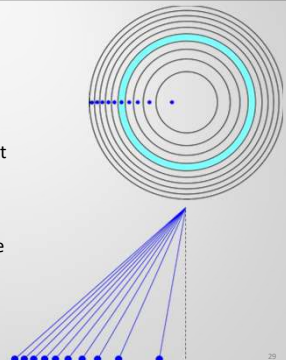
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Disc source -

The disc is treated as annuli.
16 smartly-placed samples each with a specific weighting* are used to represent the whole disc very well.

The dose rate from the disc will be the sum of the weighted dose rates from the spot samples.

*Gauss-Legendre numerical integration



Disc tests so far look OK

10 MBq ¹³⁷Cs disc source at 30 cm from the skin

Diameter	MCNP6.2 & ICRU95	95% CL	Beta Beyond 10
cm	μSv/h	μSv/h	μSv/h
20	1357	(1190-1524)	1460
40	1281	(1106-1457)	1240
60	885	(738-1033)	1030
80	709	(583-834)	823
100	607	(479-735)	664

10 MBq ³²P disc source at 30 cm from the skin

Diameter	MCNP6.2 & ICRU95	95% CL	Beta Beyond 10
cm	μSv/h	μSv/h	μSv/h
20	1434	(1373-1495)	1320
40	1246	(1189-1302)	1150
60	1064	(1011-1118)	966
80	921	(868-973)	795
100	685	(643-727)	658

10 MBq ³²P disc source at 10 cm from the skin

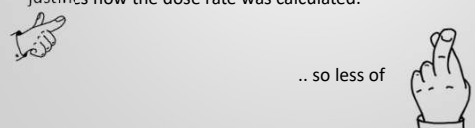
Diameter	MCNP6.2 & ICRU95	95% CL	Beta Beyond 10	Varskin+2.1
cm	μSv/h	μSv/h	μSv/h	μSv/h
10	6428	(6338-6519)	6310	4100
20	3829	(3760-3900)	4300	2500
30	2211	(2158-2254)	2800	1400
40	1318	(1277-1359)	1820	-

E: Is this of help to an RP/RPA?

Better dose estimates from foreseeable incidents
Have the usual tools under-estimated the dose rates to the extent that it challenges previous decisions?

For regulatory justification – the spreadsheet demonstrates and justifies how the dose rate was calculated.

.. so less of



Known limitations

The initial assumptions still apply.

The model was designed mainly for medium- and high-energy beta sources at distances greater than 10 cm. Results become unreliable in other circumstances.

An (optional) empirical factor is used to account for energy straggling. This is an approximation because energy straggling changes with both electron energy and travel distance, but it does give reasonably good agreement with Monte Carlo estimations.

Positrons are also in the database. I've not checked the results yet, so use with caution.

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G: Get a copy?

Github

https://github.com/RHW2026/BBT/releases/download/v1.1/BetaBeyondTen_v1_1.xlsx

Contains:
Spreadsheet designed for Microsoft Excel.

(Progressive web application TBA)



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